Title of Article

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Key Points:

- RSTF
- Source Inversion using RSTF and Teleseismic data
- Compare slip models from RSTF and Teleseismic data

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Abstract
Will be done Later.

Plain Language Summary

Will be done Later.

1 Introduction

1.1 Elastic Rebound Theory and Earthquake Cycles on Transform Faults

- Application of elastic rebound theory and seismic gap hypotheses to stress accumulation and release along fault boundaries.
- Shortcommings of the elastic rebound theory in explaining complex earthquake cycles, especially on oceanic transform faults.

1.2 Seismotectonic Setting of our study area and the fault structure

- Tectonic framework of the plate boundary, quasi-periodic seismicity, fault strike, and plate motion. (strike/dip/rake, slip rate, etc.)
- Structural characteristics of the fault plane: non-vertical, complex geometry.
- Overview of the 2015 and 2025 earthquake sequence occurring along the same fault system within a 10-year span as shown in the figure 1

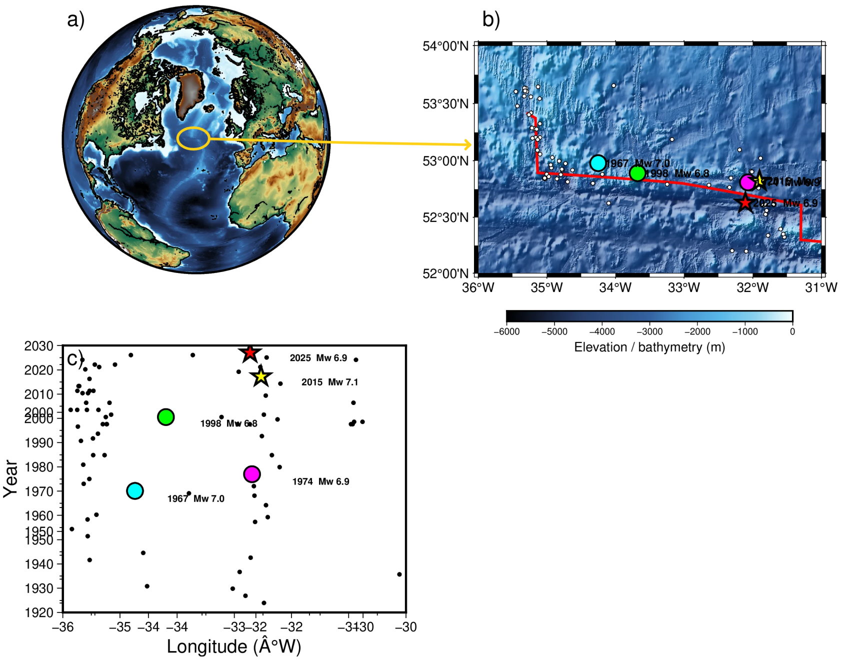

Figure 1. Seismicity and tectonic setting of the northern Charlie–Gibbs transform fault. (a) Regional tectonic setting shown on a global bathymetric map derived from the GMT Earth Relief dataset. (b) Detailed bathymetric map of the northern Charlie–Gibbs transform fault, based on the 3-arc-sec GMT Earth Relief grid (SRTM3S), with the fault trace shown as a solid line refering from (Aderhold & Abercrombie, 2016). Earthquake epicenters are from USGS ComCat, with magnitudes compiled from USGS ComCat and historical earthquake catalogues where available. Circle sizes scale with earthquake magnitude. The 1967 (M_w 7.0), 1974 (M_w 6.9), and 1998 (M_w 6.8) earthquakes are shown as colored circles, while the 2015 (M_w 7.1) and 2025 (M_w 6.9) earthquakes are shown as stars. (c) Temporal distribution of earthquakes as a function of longitude, illustrating the spatial and temporal recurrence of major earthquakes along the Charlie–Gibbs transform fault. Bathymetric data are from GMT Earth Relief; the 15-arc-sec regional grid is based on SRTM15+ V2.1 and the 3-arc-sec detailed grid is based on SRTM3S.

1.3 Unresolved Questions and Study Objectives

- As shown in Figure 1, the 2025 earthquake occurred only about 10 years after the 2015 earthquake and affected the same section of the Charlie–Gibbs transform fault. Such a short recurrence interval is not evident in the historical seismicity of the fault (Figure 1(c)). Given the relatively slow plate motion of ∼2.24 cm/yr (DeMets et al., 1990), only about 22 cm of tectonic displacement would accumulate over 10 years. This is much smaller than the several metres of slip typically associated with a large earthquake. The occurrence of two large earthquakes on the same fault segment within such a short time interval is therefore unexpected in the framework of a simple stress-loading and elastic-rebound model, where a major earthquake is expected to release accumulated strain and consequently delay another large rupture on the same segment.
- To investigate: Are the slip distributions of the 2015 and 2025 earthquakes similar or different? If different, what are the implications for stress accumulation and release along the fault system?

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2 Data and Methodology

2.1 Which methods are aplied to investigate our problem

- RSTF deconvolution from real data for inversion (Vallée, 2004).
- Teleseismic phase picking and preparation of the data for inversion.
- Finite-fault kinematic inversion using teleseismic and RSTF data to get slip distribution for both the 2015 and 2025 earthquakes (Vallée et al., 2003).
- Details of these methods will be provided in the supporting information section.

2.2 How the method is applied

2.2.1 Data Selection and Teleseismic Preprocessing for Inversion

- Continuous three-component waveform time series and corresponding metadata were retrieved for the 2015 and 2025 earthquakes from the GEOSCOPE database from the networks: IU, II, GT, G, GE, DK, CN.
- For RSTF deconvolution, we retrieved data for a radius of 40 degrees around the epicentres of the 2015 and 2025 earthquakes.
- For RSTF analysis, we selected the same earthquake which happened on 05/04/2025 as our EGF. Why this earthquake was selected as EGF will be explained in the supporting information section.
- Instrument response was also removed to maintain the homogeneity of the data.
- For teleseismic phase picking, we selected data for a radius of 80 degrees to avoid upper-mantle triplications and core reflections.
- The teleseismic P- and S-wave data were windowed for 70 and 115 s, respectively, with arrival-time uncertainties of 2.5 and 7 s. The waveforms were processed using low- and broadband smoothing periods of 33 and 8 s, respectively, and a high-frequency cutoff of 0.005 Hz was applied.

2.2.2 Finite-fault kinematic inversion

Fault Plane Geometry

2015 earthquake

The initial fault plane geometry is derived from (Aderhold & Abercrombie, 2016)

- Strike: 98°
- Initial dip: 75°
- Initial rake: 175°
- Maximum rake deviation: ±30°
- Maximum dip deviation: ±30°
- Fault length: 198 km
- Fault width: 30 km
- Fault-centre coordinates: (−77.24, 10.86, 18) km
- Hypocenter: subfault 96

2025 earthquake

- Strike: 98°
- Initial dip: 75°
- Initial rake: 175°
- Maximum rake deviation: ±30°
- Maximum dip deviation: ±30°
- Fault length: 200 km
- Fault width: 32 km
- Fault-centre coordinates: (−63.47, 8.22, 19) km
- Hypocenter: subfault 71

Fault Discretization

2015 earthquake

- Subfault dimensions: 6×6 km
- Fault dimensions: 198×30 km
- Hypocenter subfault: 96

2025 earthquake

- Subfault dimensions: 8×8 km
- Fault dimensions: 200×32 km
- Hypocenter subfault: 71

Inverted Source Parameters

The finite-fault inversion solves for the spatial and temporal characteristics of the rupture. The principal source parameters include: slip amplitude, rake, spatial variation in fault dip

Inversion Configuration

The principal numerical parameters used in the finite-fault inversion include:

- Rigidity: 4×10^{11} Pa
- Initial rupture velocity: 3.5 km/s

The complete inversion parameter configuration is provided in the Supporting Information.

Final Inversion Products

For each earthquake, the inversion provides:

- Slip distribution
- Rake distribution
- Rupture-time distribution
- Dip distribution
- Synthetic waveforms
- Observed–synthetic waveform misfit

2.2.3 Comparison of the 2015 and 2025 earthquake slip distributions

- Slip distributions are plotted for both the 2015 and 2025 earthquakes. The slip distributions are compared to understand the similarities and differences in the rupture processes of the two events.
- Contours of the slips from both earthquakes are plotted in a single figure to visualize the spatial relationship between the two slip distributions.
- To plot the contours we distributed the absolute slip values into 5 equal intervals starting from minumum slip of 8cm and plotted the contours for both earthquakes in a single figure.

2.2.4 Comparison between source time functions of the 2015 and 2025 earthquakes

- The STFs are plotted for both earthquakes and the differences in the duration, peak amplitude, and overall shape of the STFs are analyzed.

Results

Open Research Section

This section MUST contain a statement that describes where the data supporting the conclusions can be obtained. Data cannot be listed as "Available from authors" or stored solely in supporting information. Citations to archived data should be included in your reference list. Wiley will publish it as a separate section on the paper's page. Examples and complete information are here: https://www.agu.org/Publish with AGU/Publish/Author Resources/Data for Authors

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Conflict of Interest declaration

Please include a comprehensive conflict of interest statement that reflect all conflicts of interest for all involved authors. If there are no conflicts of interest please state, "The authors declare there are no conflicts of interest for this manuscript."

Acknowledgments

Enter acknowledgments here. This section is to acknowledge funding, thank colleagues, enter any secondary affiliations, and so on.

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